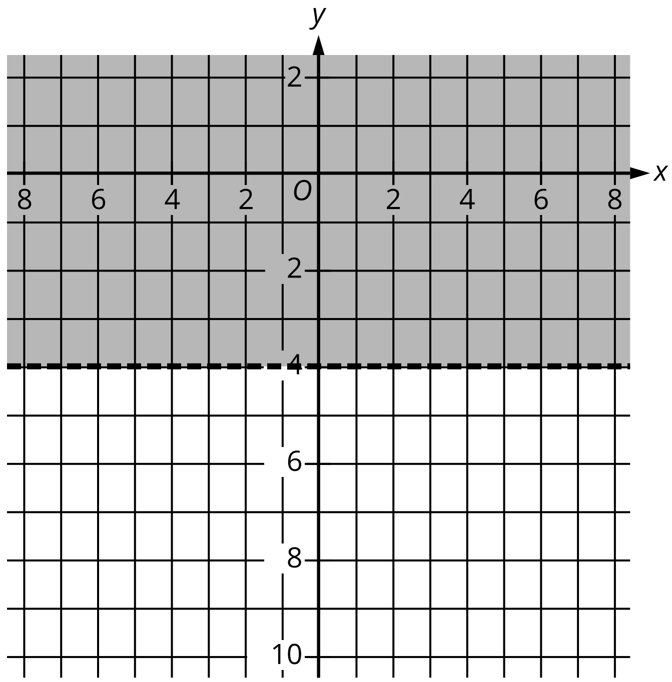
Question ID: 187f74bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question



The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r?

Answer

A. 15

B. 4

C. -4

D. -15

Correct Answer: D

Rationale

Choice D is correct. The line representing the boundary of the shaded region shown is a horizontal line; therefore, the equation of the line can be written as $y = -4$. Since the shaded region represents all of the points above this boundary line, it follows that the shaded region represents the solutions to the inequality $y > -4$. It's given that the shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. The inequality $y > -4$ can be rewritten in the form $ry < 60$ by multiplying both sides of the inequality by -15, which yields $(-15)y < 60$. Therefore, the value of r is -15.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.